```
Instructions:
1. Run task_0a.py to generate the vector database if not
already generated.
2. Press Run All (or restart kernel and run all cells).
3. You will be prompted to provide input values.
```

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- Task 1: Implement a program which, given (a) a query label,
l (b) a user selected feature space, and (c)
positive integer k, identifies and visualizes the most
relevant k images for the given label l, along with
their scores, under the selected feature space.
```

In [ ]:

```
LABEL = input("Provide a valid label name / label ID present in the Caltech1

FEATURE_SPACE = input("Provide a feature space [color, hog, avgpool, layer3,

K = int(input("Enter K, the number of similar images K to find for the selec
```

In [ ]:

```
# Find the k most similar images for a given query label for the given featu

from utils.query_input_processor import check_label_is_valid

LABEL = check_label_is_valid(LABEL)
```

```
Input label:  100
Downloading dataset if not present.
Files already downloaded and verified
Output label: yin_yang
```

In [ ]:

```
# Load feature space and representative label vectors for that feature space

from utils.database_utils import retrieve

feature_vectors = retrieve(f'{FEATURE_SPACE}.pt')
rep_label_vectors = retrieve(f'rep_label_{FEATURE_SPACE}.pt')
```

In [ ]:

```
from utils.dataset_utils import initialize_dataset
dataset = initialize_dataset()

from utils.distance_utils import get_distance_fn, top_k_min_distance_ranker
top_k = top_k_min_distance_ranker(K, rep_label_vectors[LABEL], feature_vecto

print("\nK =", K, "most similar images by", FEATURE_SPACE, "feature space us
```

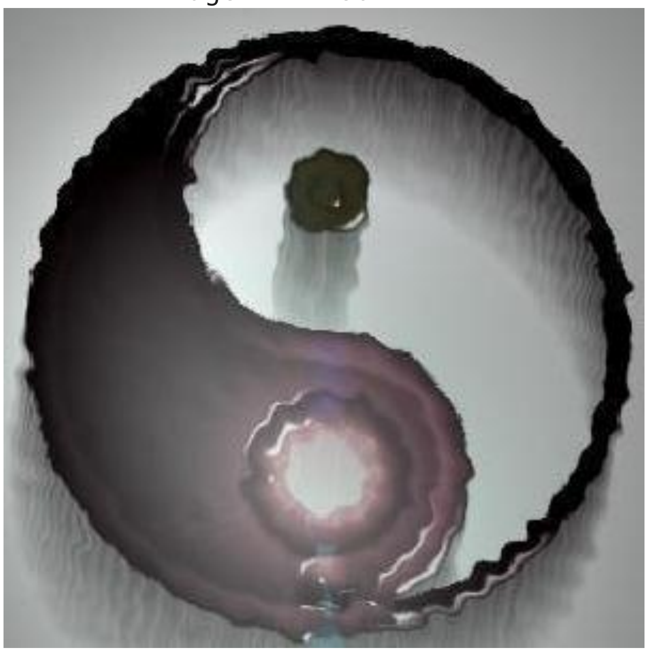
```
for i in range(len(top_k)):

    distance, image_id, label = top_k[i]

    print(str(i + 1) + ".", "\tImage ID: ", image_id, "\t\tDistance: ", dist
    display(dataset[image_id][0])
```

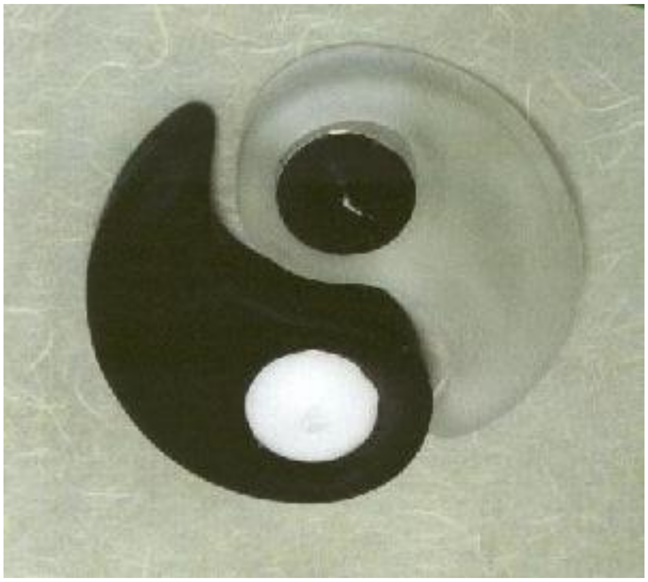
```
Downloading dataset if not present.
Files already downloaded and verified

K = 10 most similar images by fc feature space using distance: cosine
1.      Image ID:  8622             Distance:  0
```

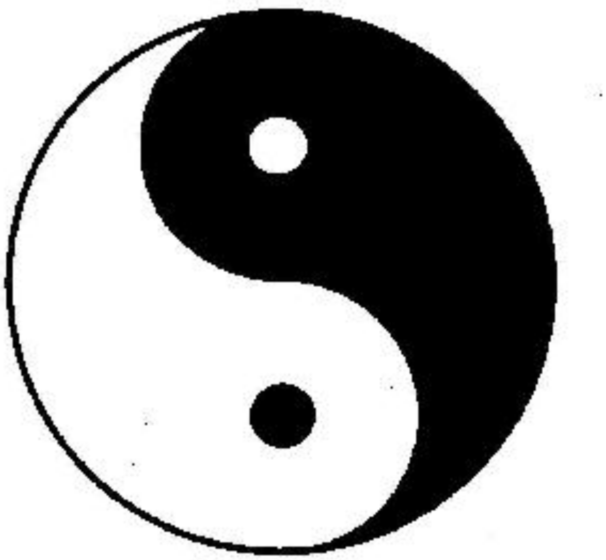


```
2.      Image ID:  8640             Distance:  0.03732657432556152
```

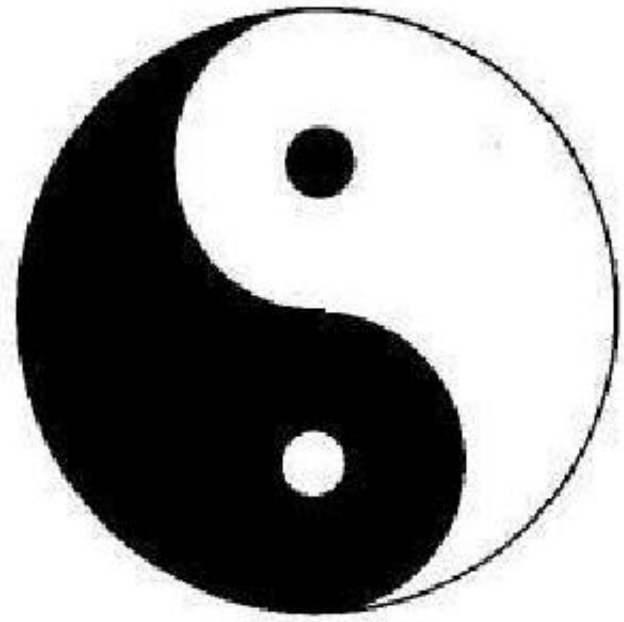
```
3.      Image ID:  8664             Distance:  0.04269057512283325
```



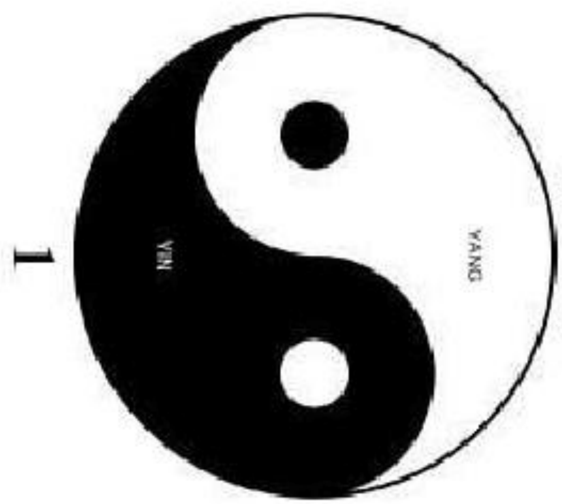
4. Image ID: 8656 Distance: 0.05204427242279053



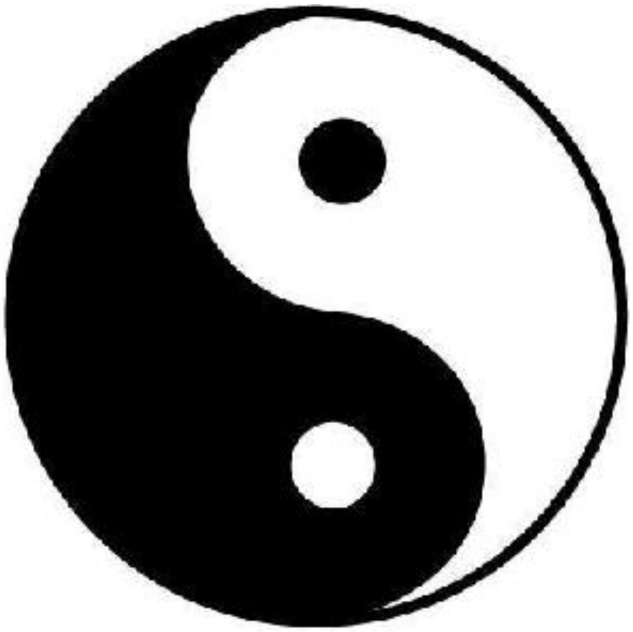
5. Image ID: 8658 Distance: 0.06078618764877319



6. Image ID: 8666 Distance: 0.06420868635177612

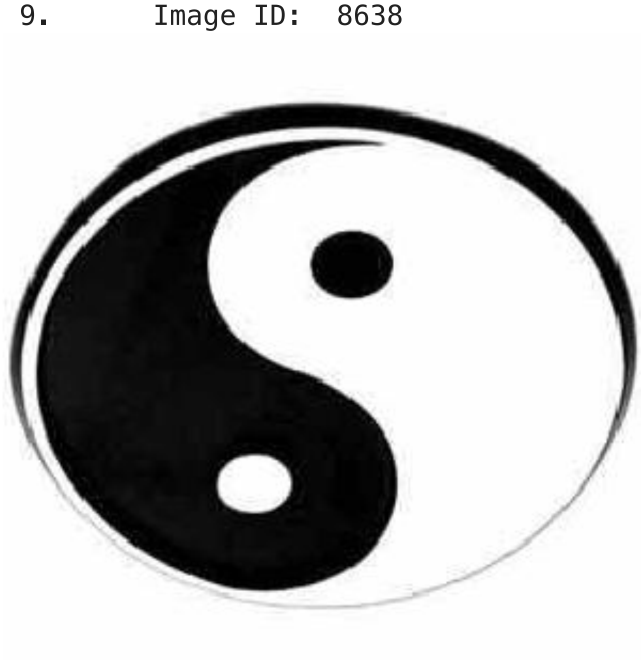


7. Image ID: 8618 Distance: 0.06979566812515259



8. Image ID: 8628 Distance: 0.07059180736541748

9. Image ID: 8638 Distance: 0.07979327440261841



10. Image ID: 8654 Distance: 0.08036291599273682